
The Limits of Audio-Only Popularity Prediction: An AutoResearch Investigation of Spotify Track Popularity

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Abstract

This paper presents a six-week autonomous research study on Spotify popularity prediction using numeric audio features. Starting from a LinearRegression baseline (RMSE 22.06, $R^2 = 0.026$), the project demonstrates that RandomForestRegressor dramatically outperforms linear models on this task (artifact-free RMSE 15.72, $R^2 = 0.548$), a 28.8% relative improvement. However, the central finding is not the performance gain but the speed at which further progress halts: 12 subsequent feature-engineering and hyperparameter experiments produced a combined 0.058-unit improvement. A popularity bucket analysis explains why. The model performs adequately on medium-popularity tracks (RMSE 13.31) but fails severely on high-popularity tracks (RMSE 28.48, systematic bias -23.26), because the factors that elevate a song to breakout popularity — artist reputation, playlist placement, and release context — are entirely absent from the numeric audio feature set. A feature-bucket error analysis confirms that this failure is localized to popularity-outcome space: slicing by danceability, energy, or tempo produces at most a 3.1-unit RMSE spread with near-zero bias, while slicing by popularity bucket produces a 15.2-unit spread and a 33.9-unit bias swing. The project’s primary contribution is characterizing this information ceiling precisely and demonstrating that it cannot be addressed through model selection, hyperparameter tuning, or preprocessing alone.

1 Introduction

Music popularity is a natural target for machine learning: it is a numeric outcome associated with structured, measurable audio properties exposed through Spotify’s audio analysis API. The intuition that a song’s energy, danceability, tempo, and acousticness should predict its popularity is plausible. The harder question is not *whether* audio features predict popularity, but *how well* and *what the ceiling is*.

This project investigates that question through a constrained, reproducible experiment. The dataset is a fixed 114,000-track Spotify corpus with 13 numeric audio features and a popularity score from 0 to 100. The evaluation metric is validation RMSE on a held-out split with a frozen random seed. The experimental protocol is a structured keep/discard/crash loop: each modification to the model or preprocessing pipeline is retained only if it improves validation RMSE. No external data, categorical features, or neural architectures are introduced.

Over six weeks and 16 experiments, this project arrives at a clear result: RandomForest with minimal log-transform preprocessing is the best achievable model from this feature set, and its accuracy plateaus sharply. The theoretical reason for the plateau is identifiable and quantifiable: the numeric audio features do not contain the information needed to predict breakout popularity. The project’s contribution is making this ceiling explicit — locating it in popularity-outcome space, ruling out

audio-feature-space explanations, and quantifying how severely it affects high-popularity track prediction.

2 Research Question

Can Spotify track popularity be predicted from numeric audio features, and what are the practical limits of audio-only prediction?

The 13 numeric features available are: danceability, energy, loudness, tempo, speechiness, acousticness, instrumentalness, liveness, valence, key, mode, time signature, and duration. The target is popularity, an integer from 0 to 100. Four sub-questions structure the analysis: (1) Do tree-based models substantially outperform linear models? (2) Do gains plateau after the initial model-family switch? (3) Does prediction quality vary by popularity range, and can the failure mode be explained? (4) Is the failure localized to the popularity dimension or to specific audio-feature profiles?

3 Methodology

3.1 Dataset

The dataset consists of 114,000 Spotify tracks. The target popularity ranges from 0 to 100 with a right-skewed distribution; breakout popular tracks (popularity > 67) constitute approximately 6.7% of the dataset ($n \approx 1,556$ in the validation split). The train/validation/test split is 60/20/20 with `random_state=42`, frozen for the duration of the project. All reported metrics are validation RMSE.

One dataset artifact was identified in Week 6: the `index` column (CSV row number, 0–113,999) was included in the raw feature set and achieved 35.8% RF feature importance because the CSV is sorted by genre, making row position a noisy genre proxy. Removing it costs 0.70 overall RMSE (15.02 $\rightarrow$ 15.72) but reduces high-popularity bucket RMSE by 3.24 units and redistributes importance evenly across the 8 core audio features. All project conclusions survive removal. The artifact-free model (RMSE 15.72) is the honest measure of what audio features alone achieve and is used as the primary result throughout.

3.2 Experimental Protocol

Each experiment modifies a single aspect of `model.py` — model family, a hyperparameter, or a preprocessing step — runs `python3 run.py`, and records the validation RMSE. The decision rule is: **Keep** if RMSE improves; **Discard** if RMSE worsens or is unchanged; **Crash** if code fails. The evaluation logic and random seed are frozen throughout, ensuring RMSE comparisons reflect genuine model changes across all 16 experiments.

3.3 AutoResearch Workflow

The project ran over five active weeks. Week 2 established the LinearRegression baseline (RMSE 22.06) and introduced RandomForest, which immediately reduced RMSE to 15.07. Week 3 confirmed RF as the retained family and discarded HistGradientBoosting (RMSE 18.45). Week 4 ran a controlled `n_estimators` sweep (`{50, 100, 200, 300}`), finding sharply diminishing returns ($6\times$ runtime for 0.5% gain) and retaining $n = 200$. Week 5 ran 12 experiments across log transforms, interaction terms, and alternative model families, retaining only `log1p(duration_ms)` and `log1p(instrumentalness)` for a combined 0.058-unit improvement. Week 6 locked the model and conducted popularity bucket analysis, feature-bucket error analysis, and index artifact validation — no model changes were made.

4 Results

4.1 Linear Baseline and Model-Family Switch

LinearRegression achieves RMSE 22.06 and $R^2 = 0.026$. The near-zero R^2 confirms that popularity and audio features are not linearly related. Ridge ($\alpha = 1.0$) produces identical RMSE, indicating the problem is structural rather than one of overfitting.

Replacing LinearRegression with RandomForestRegressor reduces RMSE from 22.06 to 15.07 — a 7.00-unit absolute improvement and 31.7% relative reduction. R^2 jumps from 0.026 to 0.545. This single architectural choice accounts for 7.00 of the project’s total 7.04-unit RMSE gain. Tree-based splitting captures nonlinear structure and feature interactions that linear models cannot represent without explicit specification.

4.2 Hyperparameter Scaling and Feature Engineering

Table 1 shows the controlled `n_estimators` sweep. Increasing from $n = 100$ to $n = 300$ costs $3.6\times$ more compute for a 0.5% RMSE gain. The bottleneck is the feature set, not model capacity; $n = 200$ is retained as the cost-performance optimum.

Table 1: Effect of `n_estimators` on validation RMSE and runtime.

<code>n_estimators</code>	RMSE	Δ vs. $n=100$	Runtime
50	15.18	+0.11	9.7 s
100	15.07	—	16.2 s
200	15.03	−0.04	32.8 s
300	14.99	−0.077	58.5 s

Table 2 shows the full ablation across all 16 experiments. The combined gain from all 12 Week 5 experiments is 0.058 RMSE units. Only `log1p` transforms on zero-inflated features (`instrumentalness`, `duration_ms`) improved the model; transforms on bounded features (`loudness`, `speechiness`, `acousticness`) worsened it. Manually engineered interaction terms (`danceability` $\times$ `energy`, `energy` $\times$ `loudness`, `tempo` $\times$ `danceability`) all worsened RMSE, since RF discovers interactions natively through tree splits. The locked final model is RF ($n = 200$) with `log1p(duration_ms)` and `log1p(instrumentalness)`.

Table 2: Full ablation across all 16 experiments.

Experiment	RMSE	Δ vs. Prior	Decision
LinearRegression (baseline)	22.06	—	Frozen baseline
Ridge ($\alpha = 1.0$)	22.06	0.000	Dropped
RandomForest ($n = 100$)	15.07	−7.00	Kept
HistGBR (default)	18.45	+3.38	Dropped
RF ($n = 200$, controlled)	15.03	−0.04	Kept
<code>log1p(duration_ms)</code>	15.02	−0.002	Kept
<code>log1p(loudness)</code>	15.03	+0.003	Dropped
Drop key, mode, time_sig	15.73	+0.711	Dropped
Interaction terms ($\times 3$)	15.04–15.06	+0.004–+0.019	Dropped
<code>log1p(instrumentalness)</code>	15.02	−0.004	Kept — final best
<code>log1p(speechiness, acousticness)</code>	15.03	+0.007	Dropped
HistGBR (tuned, 500 iter)	17.75	+2.73	Dropped
RF <code>max_features=0.5</code>	15.38	+0.357	Dropped
ExtraTreesRegressor	15.45	+0.432	Dropped
RF <code>min_samples_leaf=3</code>	15.28	+0.260	Dropped

4.3 Popularity Bucket Analysis

The strongest result of the project emerges from stratifying the validation set by popularity range (Table 3, artifact-free model).

The model performs best on medium-popularity tracks and worst on high-popularity tracks by a factor of $2.1\times$ in RMSE. The high-popularity bucket exhibits systematic underprediction of approximately 23 units — a clear regression-to-the-mean failure. The model can recognize that a high-popularity song sounds good, but cannot identify which good-sounding songs become hits, because those tracks are differentiated by unobserved factors: artist reputation, label support, editorial playlist placement, and release timing. The low-popularity bucket shows symmetric overprediction (+10.67): some

Table 3: Model performance stratified by popularity bucket (artifact-free model, RMSE 15.72).

Popularity Bucket	N	RMSE	MAE	Mean Bias
Low (0–33)	11,127	15.22	11.22	+10.67 (overpredicts)
Medium (34–66)	10,117	13.31	9.86	−8.79 (slight underprediction)
High (67–100)	1,556	28.48	23.27	−23.26 (severe underprediction)

obscure tracks sound like mainstream pop but remain unknown. The audio features signal quality but not visibility.

4.4 Feature-Bucket Error Analysis

To determine whether failures arise from misreading audio profiles or from popularity-space regression, validation errors were stratified by danceability, energy, and tempo. Across all three audio features the RMSE span is narrow: danceability buckets span 14.53–17.58 (3.1 units), energy buckets span 15.60–15.94 (0.3 units), and tempo buckets span 14.82–16.18 (1.4 units). Bias across all audio-feature slices is near zero (largest magnitude: +0.62). By contrast, slicing by popularity bucket produces a 15.2-unit RMSE span and a 33.9-unit bias swing. The model’s systematic failure does not arise from misreading audio profiles; it arises from the absence of information about which songs within any audio profile become popular. Notably, high-danceability tracks are the *hardest* to predict — the model cannot distinguish a chart-topping pop release from an obscure underground track with identical audio features.

5 Reflection

What worked. The keep/discard/crash loop ran reliably across all 16 experiments with zero crashes and correct rollbacks. Diminishing returns were identified accurately: after the Week 4 sweep showed a $6\times$ runtime increase for a 0.5% RMSE gain, the loop correctly concluded that model capacity was not the bottleneck. The log-transform criterion was applied correctly — $\log_{1p}$ helped exactly for zero-inflated features and hurt for bounded ones.

What failed. The index artifact went undetected for five weeks. It was the highest-importance feature at 35.8% the entire time, yet the agent never inspected feature importances during the experiment loop. The artifact was discovered only when Week 6 interpretability analysis began — a structural limitation of pure RMSE optimization without self-auditing. Additionally, three consecutive failed interaction experiments (E4–E6) should have been one: after the first failed, it was already clear that RF discovers interactions natively. The analysis phase started too late; bucket analysis run after Week 3 would have revealed by Week 4 that the high-popularity failure was structural, saving the resources spent on preprocessing experiments that could not address it.

What required human judgment. Three decisions exceeded what the optimization loop could make autonomously. First, the *scope lock*: stopping the model search and shifting to analysis required judgment about project framing that the agent reached the technical basis for by Week 4 but could not act on alone. Second, *artifact adjudication*: determining that all conclusions survive index removal and that the artifact-free model is the more honest result required judgment about what “valid” means for a constrained experiment. Third, the *validity constraints themselves* — the frozen evaluation logic, fixed random seed, and prohibition on external data — were human decisions encoded in `program.md` that make RMSE comparisons meaningful. An autonomous loop cannot design its own validity conditions.

6 Conclusion

This project set out to ask how well Spotify track popularity can be predicted from numeric audio features. The answer is: substantially better than a linear model, but with a hard ceiling that cannot be overcome through model refinement or preprocessing.

The progression from LinearRegression (RMSE 22.06) to the final artifact-free RandomForest model (RMSE 15.72) represents a 28.8% improvement. The switch to tree-based nonlinear splitting was the single decisive change; every subsequent experiment across four additional weeks moved RMSE by a combined 0.06 units. The feature ceiling is real and measurable.

The popularity bucket analysis locates the ceiling precisely. For medium-popularity tracks the model achieves RMSE 13.31 with near-unbiased predictions. For high-popularity tracks, RMSE is 28.48 with a systematic underprediction bias of 23 units. The factors that turn a well-made song into a breakout hit — artist reputation, playlist placement, label promotion, release timing — are entirely absent from the 13-column numeric feature set. The model cannot find what is not there.

The feature-bucket error analysis rules out the alternative that the model fails because it misreads certain audio profiles. Error is nearly flat and unbiased across danceability, energy, and tempo slices; the 15-unit RMSE gap between popularity buckets has no counterpart in audio-feature space, where the largest comparable gap is 3.1 units.

The primary contribution of this project is not the best validation RMSE. It is identifying, characterizing, and explaining the hard limit of audio-only popularity prediction: **numeric audio features describe what a song sounds like; they cannot determine which songs the world hears.** Closing this gap would require artist-level features (follower counts, prior release history), release-context features (label, editorial playlist status), and social engagement signals — structurally different from audio features in that they describe a song's position in the industry rather than its sonic character. That distinction is what the bucket analysis makes precise, and it is the natural starting point for any future work on this problem.